UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

	)	
Interconnection of Large Loads	)	Docket No. RM26-4-000
to the Interstate Transmission System	)	

REPLY COMMENTS OF EOLIAN L.P. ON THE ADVANCE NOTICE OF PROPOSED RULEMAKING

Eolian L.P. (Eolian) respectfully submits reply comments in response to the Advance Notice of Proposed Rulemaking (ANOPR) on Ensuring the Timely and Orderly Interconnection of Large Loads.[1] Attachment A to the reply comment is draft, exemplar *pro forma* language to establish a separate, streamlined study process for hybrid resources that commit to particular operating limits (i.e., to operate the combined power of a hybrid facility within maximum net withdrawal and injection limits). Eolian is a developer of energy generation and load projects, including battery energy storage systems, that develops resources, among other purposes, to serve large loads.

[1] *Ensuring the Timely and Orderly Interconnection of Large Loads*, Advance Notice of Proposed Rulemaking, Docket No. RM26-4-000 (Oct. 23, 2025) (ANOPR).

Table of Contents

I. EXECUTIVE SUMMARY 3

II. THE COMMISSION MUST ADOPT A COMPREHENSIVE SET OF LARGE LOAD INTERCONNECTION RULES TO MEET THE MOMENT 5

A. Jurisdiction 8

1. The Commission has ample jurisdiction to establish large load transmission system interconnection rates 8

2. The Commission must establish a bright-line applicability threshold for large load interconnection rules to be workable 12

B. Key elements of a rule to enable rapid, affordable interconnection of large loads and hybrids 15

1. The Commission must require transmission providers to establish a streamlined, single study, serial interconnection process for hybrid facilities 16

2. The Commission must establish a non-firm network transmission service 27

3. The Commission should ensure that large loads pay their fair share by requiring Transmission Providers to offer Transmission Security Commitments 29

III. IN THE ALTERNATIVE, THE COMMISSION MUST AT LEAST ADOPT REFORMS TO RECOGNIZE AND FACILITATE THE INTERCONNECTION AND DEVELOPMENT OF HYBRID RESOURCES 41

A. There is broad support in the record for establishment of rules to facilitate hybrid resources 42

B. The Commission has clear jurisdiction over interconnection of hybrid resources 44

C. Rules for Hybrid Resources are needed to ensure just and reasonable rates 44

D. Adopting a HILLGA-only approach would miss out on critical benefits of hybrid resources 46

E. Mandating direct assignment of network upgrade costs risks to hybrid resources risks slowing their development 48

IV. OTHER ISSUES TO CONSIDER 49

A. Option to Build 49

B. Settlement 52

C. PUHCA 53

I. EXECUTIVE SUMMARY

The Commission has the record before it to support urgent action to address the rapidly accelerating reliability and affordability crisis emerging due to large load growth. Commenters widely agree on the mounting challenges posed by large loads, and also widely agree that incentivizing flexible large loads, as envisioned through implementation of the ANOPR principles, is a key tool to mitigate those challenges. The Commission has the support on the record both to establish its jurisdiction and to demonstrate a need for reform to Commission-jurisdictional tariffs in order to unlock efficiencies in use of the transmission system, which only the Commission can achieve. The most significant pushback to action emerges on matters where the Commission can provide assurances to address concerns: first, worries that the Commission will overreach, and impact the valuable role of state oversight; and second, the potential that the Commission repeats errors of the past and establishes load interconnection processes that are as bogged down as generator interconnection queues. Both are solvable.

On the jurisdictional front, the Commission has ample scope to craft its assertion of jurisdiction so as to leave intact key state regulatory roles over retails sales and distribution facilities. It can do so broadly, and using bright-line tests based on a load's capacity and the voltage of interconnection that will provide the certainty that will be critical for investment, without impermissibly intruding on jurisdiction reserved to the states by the FPA (**Section II.A**). However, if the Commission declines to assert jurisdiction over all large loads in this proceeding, Eolian explains that it must not abandon the principles of the ANOPR to facilitate the timely and non-discriminatory interconnection of hybrid resources, where the Commission's action relies on already established jurisdiction (**Section III.B**).

On the potential for slowing down, rather than speeding up, load interconnection, the Commission can learn from experience in the generator interconnection process that upfront cost certainty for interconnection customers enables a faster process, while delaying cost certainty through after-the-fact determinations of cost liability would increase speculative proposals, withdrawals, and restudies that drain resources. On reply, Eolian builds off comments describing options for the Commission to achieve reasonable levels of up-front cost certainty and a more efficient interconnection process, while realizing the ANOPR's aim of ensuring large loads pay their fair share of the costs of the build-out necessary to reliably serve them (**Sections II.B.3**). Specifically, federal transmission security commitments, as a backstop for any comparable state requirements, can play a key role.

Eolian also underscores the strong record in support of the significant benefits of eliminating barriers to hybrid resources, and in particular the value of an expedited, streamlined study for hybrid resources that are bound to specified operating limits so as to significantly reduce their use of the bulk power system. Eolian responds to the concerns of some commenters regarding the feasibility and impacts to reliability of such study, which are largely grounded in a misunderstanding of the proposal (**Section II.B.1**). If implemented with the balanced requirements articulated in the Brattle and Elevate affidavit attached to Eolian's initial comments, which are also reflected in the model *pro forma* provisions set forth in Attachment A, an expedited, streamlined study of flexible hybrid resources provides a strong incentive, for loads that can, to invest in curtailable, dispatchable configurations at locations that minimize network upgrades and need for additional capacity, and therefore provide real impact in curbing the reliability and affordability challenges of the current moment. Indeed, if the Commission cannot yet commit to undertake the more fulsome set of reforms that are commensurate to the scale of the problem, it must at least commit to and prioritize reforms to enable hybrid resources (**Section III**). Even such more modest reforms will bring real benefit, while resting on well-established jurisdiction and having the potential to be implemented with greater speed through targeted revisions to the large generator interconnection and transmission service requirements.

Finally, Eolian responds to particular concerns raised in the docket, offering its support for workable solutions to enable the option to build for loads and hybrids; addressing the need for clear settlement rules for hybrid resources; and amplifying calls in the record for a blanket PUCHA waiver as an easy means to facilitate the interconnection and service of new large loads on the system (**Section IV**).

II. THE COMMISSION MUST ADOPT A COMPREHENSIVE SET OF LARGE LOAD INTERCONNECTION RULES TO MEET THE MOMENT

The comments on the ANOPR demonstrate that there is widespread agreement that business-as-usual is not tenable, and that significant reform is necessary to enable large load growth expeditiously, at reasonable cost, and in a manner that maintains reliability.[2] Commenters who raise questions or concerns regarding the ANOPR principles do not contest that timely

[2] NERC Comments at 2 (noting the "urgent need to properly manage the risks associated with growing demand from large load users of the BPS while also enabling them to interconnect quickly"); SPP Comments at 2 ("SPP, like other RTOs, ISOs, and utilities, is facing significant operational, planning, and market challenges in response to unprecedented growth in large, often non-traditional, load interconnection requests"); Electric Power Supply Association (EPSA) Comments at 6 ("the nation is embarking on a once-in-a-generation infrastructure buildout requiring billions of dollars of investment . . . guiding rules and provisions are needed . . . ."); American Electric Power (AEP) Comments at 2 (citing to load growth trends and arguing that "[i]t is essential for the Commission to use this opportunity to address the fundamental issues in generation interconnection, transmission planning, and cost allocation that hinder the rapid connection of large loads."); Equinix Comments at 4 ("At a time when our energy system faces its greatest need for investment in decades, efficiently integrating large loads to provide maximum benefit at optimal cost for all grid users is imperative"); Pennsylvania Public Utility Commission Comments at 2 ("Large load interconnection is rightly recognized as a matter of national importance."); Undersigned State Attorneys General and State Consumer Advocates Comments at 3-4 (recognizing "current and anticipated substantial demand growth in the near future" and emphasizing importance of a rulemaking that lessens the "ever-increasing energy cost burden"); PJM Independent Market Monitor Comments, Attachment A at 5 ("Large data center load additions have already had a significant impact and will have additional significant impacts on other customers as a result of higher transmission costs, higher energy market prices and higher capacity market prices.").

action is critical to meet the moment. Rather, such commenters question whether the Commission's assertion of jurisdiction and establishment of new requirements at the federal level can occur swiftly enough, with enough flexibility so as to not become a barrier to innovative approaches, and not result in a regime that repeats the challenges of the generator interconnection process.[3] While not diminishing the importance of each of these points, Eolian submits that the Commission has no option but to act if it seeks to ensure that load growth occurs without a catastrophic rise in energy costs and severe risk to reliability. The Commission has a window of opportunity to act before the impending crisis. The facts are stark: large loads are a national security and economic priority, but load growth cannot persist at pace under the status quo without dire impacts to reliability and affordability. Getting steel in the ground for network upgrades and new supply remains too slow, notwithstanding reform efforts.[4] If slowing the pace of load additions that will be critical to national and economic security to match the current pace of supply infrastructure development is off the table—as it should be—the only option to avoid the inevitable collision of accelerating load growth with constrained transmission and supply capacity is to ensure new loads need *less* energy infrastructure so as to be reliably integrated

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[3] Indicated PJM Transmission Owners Comments at 6-7 ("One of the key lessons learned over the last twenty years is that large, clogged queues are anathema to fast and efficient interconnections"); Pacific Gas & Electric (PG&E) Comments at 12-13 (voicing concern that existing generator interconnection queues are already under severe strain and urging the Commission to avoid new requirements that "may slow down or stall the existing approach"); PPL Electric Comments at 3 ("In some parts of the country, this assertion of jurisdiction . . . may instead slow [interconnection of crucial loads] significantly"); Chamber of Commerce Comments at 2 ("be mindful to not upend the progress currently being made . . . new delays should not inadvertently be imposed by reforms intended to achieve the opposite result.").

[4] AEP Comments at 3 (pointing to failures demonstrated by "significant queue backlogs and the repeated efforts by transmission providers to solve the problems through interconnection reforms"); DTE Electric Comments at 1 (recognizing "bottlenecks in the generator interconnection process" as one of the most acute areas of friction); Exelon Comments at 12 (notwithstanding PJM's Reliability Resource Initiative, too many projects are "stymied by other development hurdles").

faster into the system. But only the Commission has the authority to unlock these new efficiencies.

States, while essential to protecting retail ratepayer interests and critical partners in any federal regime, cannot solve the fundamentally inter-state, multi-jurisdictional problem of large load growth: only Commission oversight can enable the more efficient use and buildout of the transmission system, through reforms to interconnection processes and transmission service rules (and, where necessary, Commission-jurisdictional resource adequacy constructs). Only Commission action can eliminate barriers in federal tariffs that constrain investment in more flexible loads, which mitigate the need for new infrastructure and therefore have tremendous potential to reduce the impacts of load growth on affordability and reliability. Only by asserting jurisdiction over large load interconnection can the Commission ensure that rules reward flexibility with speed-to-power and create a meaningful incentive to develop and operate large loads differently. And only Commission action can limit the potential for undue discrimination in access to the transmission system that may otherwise result from widely divergent and too often non-standardized approaches across local jurisdictions.

The Commission must assert jurisdiction over the interconnection of large loads and establish federal requirements for the interconnections of large loads because that step is necessary to ensure that the bulk power system is capable of reliably and affordably integrating large loads. Eolian agrees with commenters arguing that any Commission action should be done in a manner that minimizes disruption of state processes and avoids a regulatory gap,[5] and that the Commission can and must learn from lessons in the generator interconnection process to

[5] Eolian Comments at 22-23 (section V: Transition to Commission Jurisdiction).

avoid implementing requirements that slow down, rather than speed up, the affordable and reliable integration of large loads. But the status quo entails unacceptable consequences: delaying integration of large loads until steel in the ground catches up or accepting rapidly escalating energy costs and mounting reliability risks as load growth persists while infrastructure build lags behind.

A. Jurisdiction

1. The Commission has ample jurisdiction to establish large load transmission system interconnection rates

The Commission not only must act, it has ample legal authority to do so. Numerous commenters urge the Commission to respect states' regulatory role over retail sales and distribution facilities.[6] The record in this proceeding establishes that the Commission has ample authority to assert jurisdiction over interconnection of large loads to the transmission system through its authority over transmission rates and, in the alternative, as a practice affecting transmission rates.[7] The record further demonstrates that the Commission can do so without asserting jurisdiction over matters that are reserved to the states.[8] As commenters explain, the Commission's regulation of this subset of large load interconnection does not disturb states' authority to regulate retail sales, including to impose terms and conditions on retail loads such as minimum monthly charges, minimum demand charges, minimum contract term, collateral requirements based on the total minimum charges for the length of the contract, load ramp

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[6] *See, e.g.*, National Association of Regulatory Utility Commissioners (NARUC) Comments at 1; Edison Electric Institute (EEI) Comments at 4; Maryland Public Service Commission Comments at 5.
[7] Eolian Comments at 5-13; Vistra Comments at 5-20; EPSA Comments at 9-13; Constellation Energy Generation, LLC (Constellation) Comments at 7-12.
[8] Constellation Comments at 12.

schedules, and mechanisms to reduce contract capacity or transfer it to another customer.[9] Nor does Commission assertion of jurisdiction over transmission-interconnected large loads usurp state regulation of distribution facilities, just as regulation of generator interconnection does not usurp state authority over generation facilities.

EEI suggests that the "most legally sound conclusion" is that Congress intended retail load interconnections to remain with the states, and points to both the savings clause in FPA section 201(a) and the prohibition against certain retail wheeling in section 212(h). EEI further suggests that the Commission's jurisdiction over interconnection—both in the context of generation and load—may be more limited than its jurisdiction over transmission, given separate statutory provisions that address interconnection. Such concerns are not persuasive. Court precedent solidly upholds the Commission's authority to regulate the terms and condition of generator interconnection notwithstanding the stark prohibition in FPA section 201(b) that the Commission "shall not have jurisdiction . . . over facilities used for the generation of electric energy," where such authority is aimed at a transmission rate, a wholesale rate, or a practice that affects either.[10] The Commission's authority to regulate large loads is similarly sound so long as, in doing so, the Commission aims at matters that are squarely in its jurisdiction. And the purposes contemplated by the ANOPR principles—ensuring efficient use of transmission and generation capacity to enable timely, affordable, reliable, and not unduly discriminatory integration of large loads—are precisely the prerogative of the Commission.

The D.C. Circuit's reasoning in *NextEra v. FERC* upholding, over objections, the Commission's jurisdiction to require replacement of equipment at a generating facility under the

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[9] Vistra Comments at 15.
[10] *NextEra Energy Res., LLC v. FERC,* 118 F.4th 361, 368-69 (D.C. Cir. 2024).

terms of a Large Generator Interconnection Agreement is instructive. Notably, this case was decided after *Loper Bright* and development of the Supreme Court's major questions doctrine jurisprudence. While the petitioners in the case argued that the Commission sought to impermissibly regulate generating facilities in contravention of FPA section 201, the Court reasoned, first, that "the action at issue directly affects facilities or transactions that FERC may regulate" because the ability to regulate the circuit breaker in question would "prevent[] other power sources from connecting to the interstate grid" which "which would directly—and substantially—limit how much electricity could be transmitted."[11] Second, the Court concluded that the Commission "had not impermissibly regulated the [power plant] as a generator."[12] Looking to its prior precedent, the Court held that the Commission may exercise jurisdiction over state-regulated facilities to the extent necessary to regulate interstate transmission, as well as where such facilities "bear a close enough relation to" an exercise of jurisdiction over jurisdictional transactions.[13] Finally, the Court concluded that the Commission's assertion of jurisdiction is "consistent with the core statutory purpose of ensuring that different generators may safely and reliably connect to the interstate transmission system."[14]

All three conclusions apply with equal force to the Commission's assertion of jurisdiction over transmission-interconnected large loads. Assuming *arguendo* that the Commission's assertion of jurisdiction reached matters that are reserved to the states under FPA section 201, the Commission's regulation of those matters is permissible for reasons that parallel the Court's findings in *NextEra*. There can be no doubt that, like generation, the terms and conditions of

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[11] *Id.,* 118 F.4th at 369.
[12] *Id*.
[13] *Id*. (quoting *TAPS*, 225 F.3d at 718 and *NARUC*, 475 F.3d at 1282).
[14] *Id*.

access of large loads to the transmission system would directly and substantially affect how and how much electricity is transmitted on the bulk power system. The addition of large loads is a key determinant of both necessary transmission build and new power supply. Second, such assertion of jurisdiction would not constitute an impermissible regulation under FPA section 201 because any incidental impact to retail loads or distribution facilities would only be to the extent necessary to regulate interstate transmission, and any impacted facilities would bear "a close enough relationship" to Commission jurisdiction over transmission facilities and transmission rates. And, as in *NextEra*, such assertion of jurisdiction would serve core statutory purposes by ensuring just and reasonable and not unduly discriminatory interstate transmission and wholesale rates while integrating large loads. The terms and conditions of interconnection of large loads certainly affect the ability of others to interconnect and broadly impact the use of the transmission system for all other users.

Nor does FPA section 212(h) change the legal analysis. The Commission would not, in establishing rules for the terms and conditions of interconnection of large loads to the transmission system, be issuing an order to wheel directly to an ultimate consumer in a manner implicating FPA section 212(h).[15]

In other words, the Commission has sufficient authority to take jurisdiction over the large-load transmission interconnection process writ large. However, if the Commission declines to assert jurisdiction over all large loads in this proceeding, Eolian argues in section III.B below that the Commission should, in the alternative, adopt rules specific to hybrid resources for whom its jurisdiction is already clearly established.

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[15] *See also* Vistra Comments at 17-19 (discussing distinction between interconnection and wheeling of power).

2. The Commission must establish a bright-line applicability threshold for large load interconnection rules to be workable

To adequately and effectively address the need for reform of large load interconnection as contemplated by the ANOPR, the Commission must establish bright-line rules as to which large loads are subject to Commission-jurisdictional interconnection processes and which are not. Bright-line applicability rules are necessary to ensure developers have sufficient certainty regarding when FERC vs. state jurisdictional interconnection procedures will apply. A clear understanding of which rules apply when will allow developers to efficiently structure facility configuration; development and investment timelines; and study processes and timelines. Bright line rules should function as "easily understandable rules, rather than complex standards, to reduce regulatory uncertainty."[16] As Advanced Energy United (AEU) explains in its comments, bright-line rules will also avoid "an otherwise unnecessary step of identifying which process [federal or state] applies to a particular interconnection request, which may introduce delay and confusion."[17] Such confusion or litigation is not merely hypothetical. AEU's comments describe a situation in which utilities within ISO-NE do not always have clear maps of which lines are FERC-jurisdictional transmission vs state-jurisdictional distribution and in which "some interconnection customers discovered very late in the process that they had been following the wrong process, through no fault of their own, adding significant delays and unnecessary costs and regulatory process."[18] The importance of bright-line rules is particularly acute for flexible loads and hybrid facilities, which may obtain certain speed-to-market benefits only by making investments that would enable them to make flexibility and curtailment commitments. Such

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[16] CAISO Comments at 4-5.
[17] AEU Comments at 10.
[18] *Id.* (citing *ISO New England, Inc.*, 180 FERC ¶ 61,129, at P 19 (2022)).

investments, which may be built into the design and financial model of a project from its earliest stages (e.g., land acquisition sufficient to site both load and proximate generation or storage facilities), will only be feasible if the developer has reasonable certainty that the benefits of flexibility as mediated by federal interconnection rules will actually be realized.

In its initial comments, Eolian proposed use of two straightforward thresholds related to capacity and voltage. Namely, the Commission could establish that loads with a capacity exceeding a given MW threshold interconnecting to facilities at a voltage exceeding a given kW threshold would be subject to federal interconnection standards. While commenters provide compelling arguments for the 20 MW large load threshold, including the potential effect loads of this size can have on the transmission system and the administrative ease for grid operators and planners in consistency,[19] others have raised reasonable concern that the 20 MW threshold proposed in the ANOPR is too low.[20] A higher threshold, such as 75-100 MW would be consistent with Eolian's project development experience. Similarly, Eolian believes that there is a range of voltage thresholds above 69 kV—some have proposed 230 kV[21]—that would reasonably support the need, articulated in the ANOPR, to ensure large load interconnection yields just and reasonable and not unduly discriminatory transmission service.

Eolian's proposed approach would apply regardless of whether a large load interconnected to a facility that was deemed "transmission" under FERC's seven-factor test. Use of the seven-factor test—which is a highly fact-intensive and too often-disputed inquiry[22]—risks frustrating the very intention of the ANOPR to ensure "timely and orderly addition of large

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[19] CAISO Comments at 5.
[20] *See* Duke Energy Comments at 15; EEI Comments at 6; U.S. Chamber of Commerce Comments at 4 (recommending that the floor for large load interconnections be raised to 75-100 MW).
[21] NRG Comments at 10.
[22] *See, e.g., Green Dev v. FERC*, 77 F.4th 997 (D.C. Cir. 2023).

loads to the transmission system."[23] While the test is longstanding, it is far from clear. As explained by NRG in its initial comments, "in a number of cases where the Commission has been tasked with applying the test, the issues have been sufficiently complex as to require haring procedures, with the result that full adjudication of the issues has taken years."[24] As a result, use of the seven-factor test would not satisfy developers' need for certainty to invest in flexible large loads and hybrid facilities and would risk inviting significant, protracted, and potentially strategic litigation.[25]

The use of clear voltage thresholds is supported by FERC precedent and by other commenters in this proceeding for similar reasons. As NRG explains in some detail in its initial comments, the need for additional certainty and uniformity was an important part of the Commission's approval of NERC's proposal to establish a bright-line presumption of 100 kV when defining the "bulk electric system" subject to federal reliability standards.[26]
As AEU explains, SPP has proposed a similar approach in its recent High Impact Large Load filing.[27]

A number of comments have recommended that the Commission adopt a voltage-based "rebuttable presumption" rather than an absolute bright-line test.[28] As explained in Eolian's initial comments, the Commission is not bound to the seven-factor test established in Order No. 888 and there are strong legal and prudential reasons to establish a set of bright-line applicability

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[23] ANOPR at P 24.
[24] NRG Comments at 7.
[25] *See* Constellation Comments at 11 ("The Commission should issue clear guidance on this point, lest the Transmission Owners use litigation over the seven-factor test to thwart the intent of the ANOPR").
[26] NRG Comments at 8-9.
[27] *Id.*
[28] *See* AEU Comments at 10; NRG Comments at 5-14 (proposing that the Commission establish a rebuttable presumption that a load interconnecting at 230 kV or above would be a large load interconnect to the transmission system).

thresholds in this instance.[29] It can, for instance, rely on its "affecting" jurisdiction under FPA section 206 to establish federal requirements for loads that exceed a particular size that are interconnecting above a certain voltage without disturbing the use of the seven-factor test for determining which facilities are transmission and which are distribution. The Commission could clearly do so in a manner that would satisfy the U.S. Supreme Court's two-step test in *EPSA*:[30] (1) the regulation of such interconnection would "directly affect" wholesale and transmission rates that fall squarely within the Commission's jurisdiction, and (2) the FERC-regulated interconnection process requirements would avoid directly regulating matters within the states' exclusive jurisdiction including retail sales and distribution facilities by aiming only at the transmission-related elements of the large-load interconnection such as transmission studies, and the allocation of the cost of transmission facilities (interconnection facilities and network upgrades). However, to the extent the Commission decides to apply new federal interconnection requirements only to those loads directly interconnecting with transmission facilities, and to continue to rely on the seven-factor test in drawing the line between transmission and distribution, Eolian supports use of a rebuttable presumption such as those proposed by AEU and NRG to provide some additional clarity and certainty.

B. Key elements of a rule to enable rapid, affordable interconnection of large loads and hybrids

To seize upon the potential benefits that flexible and hybrid facilities can provide to both large loads and existing customers, the Commission must establish rules to enable a streamlined serial interconnection process for flexible large loads and hybrid facilities that agree to be

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[29] Eolian Comments at 9-13.
[30] *FERC v. Elec. Power Supply Ass'n*, 577 U.S. 260 (2016).

curtailable and dispatchable, as applicable. Relying on such a study process, which provides for a single study of both withdrawal and injection by hybrid facilities to facilitate granting of both withdrawal and injection rights, is technically feasible and consistent with existing models that require compliance with operating limits. In order to establish an operational framework for curtailment commitments, the Commission should also establish a non-firm network transmission service class that allows loads and hybrids to withdraw from and inject into the grid during times of available transmission system capacity but not when transmission facilities are stressed or system resources are inadequate to serve the facility. Last, the Commission should ensure that large loads pay their fair share of transmission system upgrades required to serve them through Transmission Security Commitments paired with rolled-in pricing to provide cost certainty to developers while protecting existing customers.

1. The Commission must require transmission providers to establish a streamlined, single study, serial interconnection process for hybrid facilities

a. Integrated Study

Consistent with ANOPR Principles Three, Five, and Six, integrated joint study of hybrid load and generation facilities based on requested withdrawal and injection rights is critical to enabling rapid, affordable interconnection of large loads and hybrids. Some commenters question whether such integrated study can be implemented and relied upon. These commenters misunderstand the nature of the ANOPR and Eolian's proposals for integrated study of hybrid facilities. Rather, integrated joint study based on requested withdrawal and injection rights is practical and technically appropriate. Similarly, hybrid facilities can be readily configured to comply with operating limits. Given the workability of integrated joint study of hybrid facilities accounting for operating limits, the Commission should define hybrid facilities based on clear technical standards that allow for flexibility in commercial configurations.

i. Jointly studying generation and load is practical and technically appropriate

The technical affidavit attached to Eolian's initial ANOPR comments (Brattle and Elevate Affidavit) proposed an expedited interconnection process for hybrid facilities that studies the generation and load of a hybrid facility together.[31] This is consistent with ANOPR Principle Three, which provides that "to the extent practicable, load and hybrid facilities should be studied together with generating facilities" to "allow for efficient siting of loads and generating facilities and thereby minimize the need for costly network upgrades."[32]

Some commenters have raised concerns regarding Principle Three. These comments highlight the opportunity to clarify that studying facilities together (as if they both exist) does not mean merging them into a single facility and only studying the load net of generation or vice versa. Rather, it means assessing the components of a hybrid facility together in the same study construct, recognizing that they are separate components. The separate hybrid facility components are thereby treated the same as all the other facilities that exist in the transmission network except that they are subject to enforceable operational commitments. While certain contingency scenarios will recognize system protection schemes that dynamically link the output of the hybrid facility components, this is no different than similar schemes that exist today.

[31] Eolian Comments, Attach. A Levitt et al, Technical Affidavit on behalf of Eolian Energy (Brattle and Elevate Affidavit).

[32] ANOPR at P 20. The ANOPR goes on to explains that "siting a large load near or at the same point of interconnection as a new generating facility could reduce the network upgrades needed to interconnect only the load or only the generating facility." *Id.*

Constellation, for example, states that "[i]t is our understanding that a single, combined study is not practicable, as the generation and large loads typically still must be studied separately in case the load or generator trips, and either the generator injects its entire output onto the grid, or the load withdraws its full demand from the grid."[33] This statement reveals two misconceptions about the ANOPR's proposal and proposals like our own: first, that joint studies would not account for potential contingencies within a hybrid facility in the ordinary course of the study process and second, that all of the generation units within a hybrid facility would necessarily be subject to a common failure mode.

As to the first misconception: ordinary contingency studies simulate system conditions after any one (or sometimes two) facility on the transmission system is removed from service due to a contingency event. Every facility on the transmission system is treated this way in an array of "what-if" scenarios based on substation configurations to identify all possible loss of transmission components. The load and generation components of a hybrid facility are no different and would be separately subject to such contingency studies under a joint study approach.

As to the second misconception: many hybrid facilities will consist of multiple generation units (e.g., Bloom Energy's individual fuel cells or Volta Grid's individual small gas turbine units) connected to the load through redundant transmission lines, such that no single point of failure would cause all generation units to trip offline simultaneously. A similar configuration is possible for multiple redundant load modules that will not all trip offline simultaneously. In such a case, there is no contingency event in which all generation or all load trips offline at once.

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[33] Constellation Comments at 13-14.

Thus, studying the cases in which all load withdraws from the grid or all generation injects (unless requested by the hybrid facility in order to benefit from maximum as-available transactions) would not reflect realistic contingencies, and would likely result in unnecessary network upgrades. This highlights the need to define realistic contingency scenarios for hybrid facilities and their redundant configurations of generation and load components.

While it is possible that in some cases there may be operational risks that generation or load may trip, causing a sudden change in net power exchange with the grid that requires an immediate control response, the ANOPR addresses this concern in Principles Five and Six regarding requested operating limits and the need for system protection facilities respectively.[34] We also discuss this further in the following section.

ii. Operating limits can be effectively enforced through physical or other means, and are therefore appropriately incorporated into studies

ANOPR Principle Five suggests that "hybrid facilities should be studied based on the amount of injection and/or withdrawal rights requested."[35] The Brattle and Elevate Affidavit supported this principle noting that "the hybrid facility would nominate—possibly in coordination with the transmission owner—operating limits (in MW) for aggregate net injections and withdrawals of power (consistent with ANOPR principle five). These represent the maximum levels of net injections and withdrawal under which the grid can be safely operated without material upgrades."[36]

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[34] ANOPR at PP 22-23.
[35] *Id.* at P 22. The ANOPR continues that, "[f]or example, a hybrid facility consisting of a 500 MW load and a 600 MW generating facility may seek no withdrawal rights and 100MW of injection rights. This provides incentives for co-location with new generation facilities and ensures efficient buildout of the transmission system." *Id.*
[36] Brattle and Elevate Affidavit at 12.

Some commenters have raised concerns about the feasibility of implementing ANOPR Principle Five. For example, PG&E suggests that joint operating limits are problematic because it is not realistic to assume that a hybrid facility will honor them.[37] We agree with PG&E that some hybrid facilities will expect to run their generation components only occasionally, in which case they must be studied for operating limits reflecting full load withdrawals, allowing them to operate reliably under the proposed hybrid facility construct. However, other hybrid facilities will use (and already have) a bank of generation units to serve the load in a baseload configuration and will never withdraw the full load from the grid. The latter configuration may reasonably request a zero MW operating limit on withdrawals. We therefore disagree that operating limits are categorically unrealistic. Rather, operating limits can be relied upon if the regulatory context provides sufficient incentives to implement configurations that can comply with them. Operating limits can be enforced through physical system protection and control equipment, through contractual and legal tariff obligations to obey dispatch instructions, and through FERC and NERC enforcement actions.[38] This enforcement regime will incent hybrid facility interconnection customer to configure their system to comply with operating limits; if they submit their configuration and request the appropriate operating limits, the transmission provider should study them accordingly.

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[37] PG&E Comments at 13-14 ("The ANOPR proposes that hybrid facilities be studied based on the amount of injection and/or withdrawal rights required and provides an example of a 500 MW load and 600 MW generator that are co-located seeking no withdrawal rights and 100 MW of injection rights. This aspect of the ANOPR is problematic and requires modification. This proposal does not account for the likely situation that for any given time period, a co-located generator may be off-line (requiring the load to solely utilize the transmission system) or a large load facility may be partially or completely off-line or using less energy than forecasted (resulting in the generator injecting more into the transmission").

[38] Notably, CAISO explains that its "experience with large generators demonstrates that specific penalties for unauthorized injections are unnecessary" because "[s]ystem protection facilities should physically prevent injections or withdrawals that substantially impact the transmission system, including circuit breakers that would isolate the facility from the grid before it could affect reliability." CAISO Comments at 11-12.

Some parties view protection schemes differently for injections versus withdrawals. For example*,* SPP rejects operating limits for net withdrawals, stating that "protection schemes themselves may fail under fast dynamic conditions" and "very rapid withdrawals of large amounts of power can adversely affect the transmission grid," even while they recognize operating limits on net injections.[39]

Several other parties, however, support or have existing examples to demonstrate that enforceable operating limits can be made workable. First, ANOPR Principle Six already proposes that system protection facilities be installed to prevent unauthorized injections or withdrawals.[40] As mentioned above, SPP supports studies that honor operating limits on net injections (while recognizing that a customer may request the ability inject at full generator nameplate).[41] CAISO supports relying on system protection facilities to prevent operations beyond the requested withdrawal/injection rights discussed in Principle Five, explaining that "CAISO generally supports this proposal so long as it remains strongly coupled with principle six to maintain reliability. Hybrid facilities that have strict, unwavering protections like circuit breakers at their point of interconnection should be allowed to have network upgrades based on their net rights requested. This will save ratepayers and developers from unnecessary costs and help to avoid long construction timelines."[42]

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[39] SPP Comments at 10–11.
[40] ANOPR at P 23 ("Sixth, any hybrid interconnection shall be required to install the system protection facilities necessary to prevent unauthorized injections or withdrawals that exceed the respective rights. We seek comment on whether other operational limitations should be considered. We also seek comment on the minimum technical requirements for such system protection facilities, whether a hybrid interconnection customer should be subject to penalties for unauthorized injections or withdrawals, how any such penalties should be designed, and how such penalties should be allocated to other transmission customers.").
[41] SPP Comments at 11.
[42] CAISO Comments at 8.

Second, experience with system protection facilities and operational curtailment to stay within requested withdrawal/injection rights is not new. American Transmission Company (ATC) points to the example of paper mills that typically have on-site generation as an important precedent for how paired generation and load facilities operating within a specified envelope can function: "Typically, the paper mills provide ATC with load forecasts of the net amount of usage, in other words, the total load the paper mill necessitates to operate and function, minus the amount they rely on from their hydropower generation. These co-located generators and loads are 'process-matching,' meaning that if the co-located hydropower generation failed, the paper mill would cease operations. This failsafe allows ATC to minimize reliability risks associated with the paper mills providing 'net' amounts of load in forecasts."[43]

Third, MISO's surplus interconnection service program provides another example of how operating limits can be reasonably relied on. MISO relies on agreements between entities sharing generator interconnection capability to ensure that their combined injection on to the grid does not exceed the approved interconnection limit, even if their nameplate capabilities might enable them to do so. This is done through a control scheme and additional agreements that limit injections at the point of interconnection limit.[44]

Taken together, these examples clearly demonstrate that operating limits on hybrid facilities' net injections and withdrawals to and from the grid can be safely enforced, and therefore assigned based on the amounts requested, and not necessarily their nameplate

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[43] ATC Comments at 11.
[44] MISO, Generator Interconnection Business Practices Manual No. 015, BPM-015-r26, at 114 (Aug. 2, 2023), https://cdn.misoenergy.org/20230918%20PAC%20Item%2002c%20BPM-015%20Generator%20Interconnection%20Queue%20Reform%20Redlines630228.pdf.

capabilities, in such cases where such requested injections and withdrawals are below the amount of nameplate generation and load respectively.

iii. Studying generation and load based on their requested withdrawal and injection rights is technically appropriate

Nevertheless, Entergy Operating Companies (Entergy) questions the reliability of operating limits even in the presence of system protection facilities, arguing that "[e]ven if such system protection facilities could work perfectly, it is a mistake to assume that a net injection of 100 MW (where the actual withdrawals are not studied) has the same effect on the transmission system as a 100 MW generator (with no actual load). In other words, netting should not affect the way an interconnection request is studied. Studying hybrid facilities in the manner contemplated by the proposed ANOPR may fail to identify needed network upgrades, because it will be blind to potentially substantial new loads, as long as those new loads are 'hybrid.'"[45]

We agree with Entergy that it is incorrect to assume that a net injection of 100 MW is equivalent to a 100 MW generator on the system. However, the ANOPR does not imply that they are the same or that hybrid facilities should reduce reliability scrutiny by relying on netting for the purpose of interconnection studies. As described above, we believe this to be a conceptual misunderstanding of the ANOPR proposal. Joint studies do not imply only studying the load net of generation or vice versa. Rather, it means studying both the generation and load *together*, and not ignoring the presence of one while studying the other. As a result of studying them together, it may be that the net power flow onto the broader transmission system in the base case is expected to be 100 MW as suggested in the ANOPR's proposed example with 500 MW of load

[45] Entergy Comments at 27.

and 600 MW of generation in a hybrid facility. But the point is that hybrid generation and load should be studied together and considering their requested operating limits, not merely that they should be studied according to the crude difference between their total capacity. Such modeling also effectively captures all stability and short-circuit impacts, which are typically studied for full nameplate capacity of the facility.

As stated in the Brattle and Elevate Affidavit, the hybrid facility can select its chosen operating limit for net withdrawals and injections and be obligated to only operate within those limits.[46] Appropriate system protection facilities or other protocols are capable of ensuring those limits are respected.

iv. Hybrid facilities should be defined based on clear technical standards that allow for flexibility in commercial configurations

ANOPR Principle 3 explains that "siting a large load near or at the same point of interconnection as a new generating facility could reduce the network upgrades needed to interconnect only the load or only the generating facility."[47]

Duke Energy Corporation (Duke Energy) questions the feasibility of this principle too, arguing that "'[n]earness" of load and generation is not a technically implementable standard, and the 'nearness' of load and generation does not determine the electrical impacts to the transmission system."[48] Rather, Duke Energy recommends that, "[i]f the Commission seeks to establish standardized procedures for coordinating load and generation studies, including for hybrid facilities . . . any definition of 'hybrid facilities' require that the load and generation be behind a common point of interconnection to the transmission system, although the transmission

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[46] Brattle and Elevate Affidavit at 10.
[47] ANOPR at P 20.
[48] Duke Energy Comments at 18.

provider should be able to require metering and telemetry to separately monitor the load and generation behind the common point of interconnection."[49]

We agree with Duke Energy that impacts on the transmission system are the relevant consideration for any standard and recommend a configuration-neutral standard that allows parties to flexibly meet the required de-minimis impacts on the transmission system. The Brattle and Elevate Affidavit proposed a technically implementable standard for nearness that would "allow for only 3% of self-supply power to flow through ties connecting the local area to the network, and only 1% to flow over other transmission facilities outside the local area."[50] This standard achieves the goal of limiting transmission system impacts while remaining neutral on the required configuration. A standard that requires locating behind the same point of interconnection, as Duke Energy suggests, may leave many projects unworkable due to constraints unrelated to the transmission system such as acquisition of adjacent land parcels, state retail laws, and more. This would ultimately undermine the goals of the ANOPR.

Instead, the standard proposed in the Brattle and Elevate Affidavit would set clear technical requirements while allowing variations in the configuration depending on project circumstances. We urge the Commission to set standards that allow for configurations to flexibly navigate commercial and local regulatory considerations while adhering to demonstrable de-minimis impacts on the transmission system and thereby qualifying for expedited interconnection.

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[49] *Id.*
[50] Brattle and Elevate Affidavit at 8.

b. Expedited Interconnection

A large majority of commenters in this docket support the concept of expediting interconnection for curtailable and dispatchable hybrid facilities. Supportive commenters include large load customers,[51] generation and storage developers,[52] utilities,[53] and state entities.[54] Even comments that at first glance may be construed as opposing expedited interconnection support an approach that rewards more limited service requests with the benefits of streamlined study that encapsulates the more limited nature of the service requests.[55] In Eolian's view, just as an express checkout line appropriately rewards those whose needs can be addressed more quickly with faster service, it is *appropriate* to study curtailable and dispatchable facilities on a separate, streamlined basis due to their lesser impacts on the broader system (and therefore on other

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[51] *See, e.g.*, Microsoft, Inc. (Microsoft) Comments at 10 ("[R]egulatory frameworks that allow for voluntarily curtailable loads make sense and will be an important tool for accelerating the deployment of critical AI infrastructure."); Google LLC Comments at 3-4("[T]he Commission should prioritize reforms that will expedite the interconnection of facilities that maximize the use of existing infrastructure and minimize strain on the grid.").

[52] *See, e.g.*, EDF Power Solutions, Inc. (EDF) Comments at 16 ("The Commission should fast-track curtailable loads and curtailable/dispatchable hybrid facilities with firm expedited timelines."); Constellation Comments at 23 (agreeing with Principle 7 and offering "additional ways to advance curtailable and dispatchable large loads and hybrid facilities along with expediting their interconnection studies that the Commission should seek to encourage"); EPSA Comments at 16 ("EPSA firmly agrees that, when a large load customer agrees to be curtailable and dispatchable in order to assist in system reliability and flexibility in defined circumstances, that load should be subject to an expedited interconnection study process.").

[53] *See, e.g.*, Alliant Energy Corporate Services Comments at 13 ("Curtailable Large Loads Should Receive Expedited Study Treatment and Flexibility in Curtailment Agreements."); Tri-State Generation and Transmission Association, Inc. (Tri-State) Comments at 7-8 (urging the Commission to require that "*all* large loads and hybrid facilities should be curtailable/dispatchable until any required network upgrades have been constructed," and indicating that "[t]o the extent a large load agrees to be treated that way indefinitely, Tri-State does not oppose expediting such loads.") (emphasis added).

[54] *See, e.g.*, Undersigned Attorneys General and State Consumer Advocates Comments at 4, 8 ("The State Entities . . . encourage the Commission to pursue and prioritize the ANOPR principle that seeks to incentivize large loads "to be curtailable and dispatchable" by a "system operator."); Maryland Energy Administration Comments at 5 (stating, in response to the ANOPR's seventh principle, that "MEA believes well-designed voluntary pathways to faster interconnection can provide effective incentives").

[55] That is, such commentors oppose priority *access* to the queue but may not oppose tailoring study practice to the nature of the interconnection request that results in faster processing in practice. For example, Transmission Access Policy Study Group (TAPS) states that "Qualifying Large Load and Hybrid Facilities should *not* be permitted to cut the queue and receive expedited study treatment if they agree to be curtailable and dispatchable." TAPS Comments at 17-18. However, they clarify that "it may be appropriate to study Qualifying Large Load and Hybrid Facilities that agree to being curtailable and dispatchable differently in the interconnection process." *Id.* at 20-21.

resources seeking to interconnect to that system). Moreover, the rationale for streamlined study is even stronger in this context because curtailable and dispatchable facilities can be interconnected without causing a need for restudies in the broader queue.[56]

Moreover, the record supports the ability of transmission providers to study curtailable and dispatchable service requests on an expedited basis. While many commenters pushed back against the proposed 60-day study deadline, some leading transmission providers highlighted the feasibility of similar study timelines. For example, Southwest Power Pool notes that its High Impact Large Load and related High Impact Large Load Generator Assessment will complete studies of both load and generation facilities in 90-day study timeframes.[57]

2. The Commission must establish a non-firm network transmission service

The establishment of a non-firm network transmission service tier should be a core part of any final rule, as such a service tier is integral to enabling transmission providers to expedite interconnection for curtailable large loads and curtailable and dispatchable hybrid facilities. An overwhelming number of commenters endorse the concept that the interconnection of curtailable large loads and curtailable and dispatchable hybrid facilities should be expedited.[58] And numerous commenters explain that implementing this principle in practice requires the specification of non-firm withdrawal rights.[59] Indeed, some of the limited opposition registered

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[56] *See* Brattle and Elevate Affidavit at 4 (noting that such streamlined study would take place outside the standard interconnection queue and be designed to avoid the need for network upgrades).
[57] SPP Comments at 5.
[58] Advanced Energy Management Alliance (AEMA) tallies more than a hundred supportive comments as compared to a handful of cautionary voices. *See* AEMA Reply Comments at 1.
[59] *See* Thermal Battery Alliance Comments at 5 ("To facilitate the streamlined interconnection study process described above, and provide for just and reasonable transmission rates, transmission providers should be required to offer non-firm transmission service appropriate for loads that have committed to avoid specific constraints.");

against expedited interconnection for curtailable and dispatchable large load customers and hybrid facilities stems from an assertion that such curtailment is incompatible with network integration transmission service rights that may be held by such customers in the status quo regulatory framework.[60] Because the *pro forma* open access transmission tariff does not currently provide for a non-firm network transmission service offering, the Commission should require transmission providers to offer such a service as part of its implementation of principle 7.

As emphasized in Eolian's initial comments, such service should be offered on both a permanent and temporary basis, with the temporary service offering supporting "bridge" solutions that bring large load customers and hybrid facilities online while requisite network upgrades or generation resources are being constructed to enable fully-firm network integrated transmission service.[61] Southwest Power Pool is leading in the creation of both types of services.[62]

The pricing for such service should depend on whether the customer plans to take non-firm service on a permanent or temporary basis. Where the facility seeks non-firm service on only a temporary basis as network upgrades are constructed, it may be appropriate for pricing to reflect the fact that the customer will ultimately require network upgrades in the same manner as

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AEMA Comments at 6 (calling for the development of a "Non-Firm Withdrawal Right"); GridStor Comments at 7 (calling for the creation of non-firm or conditional-firm transmission service offerings); Constellation Comments at 23-25 (explaining the importance of providing for an interruptible network transmission service offering, which is not providing for in current tariffs).

[60] *See* ATC Comments at 15-16 (explaining that "pursuant to MISO's Tariff, all network load is generally treated as firm service. . . . Because ATC treats all network load as firm, even if a hybrid facility volunteers to be curtailable, ATC operators may not be able to selectively curtail that facility ahead of other network load.").

[61] Many commenters support providing for "bridge" solutions to facilitate speed-to-power. *See infra* n.83.

[62] *See* SPP Comments at 6 (describing its efforts to build temporary and permanent non-firm service options, including Conditional High Impact Large Load Service (CHILLS)).

other customers. The Commission could draw from SPP's CHILL example in providing for the creation of such temporary service offerings.

However, when a customer seeks non-firm service on a more permanent basis, pricing should reward the customer for the fact that, because it is curtailable and, for hybrids, dispatchable, the interconnection customer does not create the same system needs. In taking swift action to require the establishment of non-firm transmission service structures, the Commission could articulate a broad requirement (e.g. the transmission provider must demonstrate that its pricing for non-firm transmission service is just and reasonable), or provide more specific requirements or even *pro forma* tariff language. The Commission could draw from existing pricing of similar products. For example, Secondary Service provides customers transmission "on an as-available basis, at no additional charge."[63] Alternatively, the Commission could model scheduling and charges off of the *pro forma* OATT's non-firm point-to-point service product. Another option would be to provide for transmission providers to calculate a standby service charge for such service, as recommended by Zachary Ming of Energy and Environmental Economics, Inc., in support of Constellation in the Commission's Order-to-Show cause proceeding concerning co-located generation and large load customers in PJM.[64]

3. The Commission should ensure that large loads pay their fair share by requiring Transmission Providers to offer Transmission Security Commitments

The ANOPR states that "load and hybrid facilities should be responsible for 100% of the network upgrades that they are assigned through interconnection studies."[65] While Eolian agrees

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[63] *See pro forma* OATT, section 28.4.

[64] *See* Constellation, Answer, Docket Nos. AD24-11-000, EL25-49-000, EL25-20-000, Exhibit 2, Testimony of Zachary Ming at 27-42 (filed Apr. 23, 2025).

[65] ANOPR at P 25.

with the policy rationale behind this recommendation—the importance of ensuring that other transmission customers are protected from the costs associated with building network upgrades to serve large load customers and hybrid facilities—the record provides a better path to achieving this goal: Transmission Security Commitments to ensure that large load customers, in aggregate, will contribute their fair share to network upgrade costs via rolled-in pricing. Such an approach, when combined with the availability of non-firm transmission service (with differentiated rolled-in pricing for Transmission Security Commitments), will maintain the appropriate incentives for flexibility that are critical to achieving the ANOPR's objectives and to ensuring just and reasonable and not unduly discriminatory rates for such flexible loads and hybrid resources. Moreover, this approach would address the central source of many commenters' opposition to FERC action: the risk that the establishment of a FERC-jurisdictional load interconnection process could slow down rather than speed up the interconnection of large load customers. FERC could avoid repeating the problems that have plagued generator interconnection queues through an approach that provides greater up-front cost certainty to interconnection customers, which is the single most powerful tool for reducing speculative and duplicative requests, study complexity, and re-studies. And by requiring Transmission Security Commitments, FERC could simultaneously ensure that large load customers pay their fair share of system costs.

The Commission is correct to be concerned with protecting existing customers against shouldering the costs of network upgrades associated with large load interconnections. Exelon notes that "[o]verstated demand expectations can, and likely will, trigger costly system upgrades or new facility construction to serve that anticipated demand[,]" and that "[i]f expected demand never materializes, infrastructure and capacity costs can end up borne by other customers."[66] According to the National Rural Electric Cooperative Association (NRECA), "[t]o meet aggressive timetables, utilities are being asked to take unsecured risks, putting native-load customers at risk of covering costs if the projects do not come to fruition."[67] Tri-State states that the potential for large loads to "cease service before there is an opportunity to recover" the cost of the "significant network upgrades" that they trigger is the biggest financial risk caused by large load growth.[68] While direct cost assignment is one way to address uncertainty associated with large load customers, requiring Transmission Security Commitments of appropriate form and tenor would also protect existing customers from having to shoulder the cost of network upgrades caused but never recovered from large load customers where the load does not materialize as expected. And it can better do so while achieving the ANOPR's objectives of "timely, orderly, and non-discriminatory" interconnection of large loads and by maintaining appropriate incentives for flexibility.

a. A Transmission Security Commitment could be designed to protect existing transmission customers

In lieu of direct assignment, the Commission could provide for a Transmission Security Commitment requirement to ensure that existing transmission customers are protected. Such a requirement would build from an approach taken by Exelon and other utilities. Exelon's comments summarize the approach Exelon has taken to "protect both existing and new customers from the cost ramifications of speculative or inflated large load interconnection

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[66] Exelon Comments at 3.
[67] NRECA Comments at 6.
[68] Tri-State Comments at 6.

requests, or those that do not meet their load projections."[69] Exelon uses "Transmission Security Agreements" requiring "that a new large load customer guarantee its annual payments for transmission service will align with the" large load customer's "expected annual payments . . . for its first ten years of service."[70]

This requirement aims "to weed out speculative customer behavior while protecting other customers in the event a large load project realizes less than its promised load."[71] It is designed to be "sufficiently large, and graduated, so as to discourage speculative behavior," without "chill[ing] interest in development of real and committed large load projects."[72]

To preserve flexibility and respect jurisdictional boundaries, while simultaneously ensuring that the security required is adequate, the Commission could require Transmission Providers to establish rules providing that the large load customer or hybrid facility provide a Transmission Security Commitment sufficiently large so as to protect existing customers and that is roughly proportionate to the risk posed by the large load, without being so large as to chill investment in new load.[73] Eolian takes no position as to whether the particular contractual commitments required by Exelon strike the correct balance, but section II.B.33.c below discusses various options the Commission has to ensure that an appropriate amount of security is gathered.

In so doing, the Commission could specify that Transmission Providers should provide that state-jurisdictional security agreements or other take-or-pay arrangements pursuant to state-jurisdictional tariffs may be used as evidence that the Transmission Security Commitment has

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[69] Exelon Comments at 4.
[70] *Id.*
[71] *Id.* at 5.
[72] *Id.*
[73] As set forth in section II.B.3.c below, the Commission may wish to prescribe more specific requirements to ensure that the Transmission Security Commitment is sized appropriately to cover network upgrades caused by large load customers and hybrid facilities in aggregate.

been met. This would recognize that while the Transmission Security Commitment is a distinct federal requirement which does not impose any rules or requirements on what security may be required by a retail customer, the provision of such security at the retail level may have relevance as to whether the *Transmission Provider* can be reasonably certain that it will recover *transmission costs* sufficient to pay for network upgrades needed to serve its large load customers. Indeed, Exelon notes that its own Transmission Security Agreement structure is "buttressed by expanded deposit requirements for large customers at the retail level through state-jurisdictional service request processes."[74] For example, the Indicated PJM Transmission Owners note that the Public Utilities Commission of Ohio and Indiana Utility Regulatory Commission each recently accepted tariff revisions to protect customers from large load additions: by AEP-Ohio to create a data-center specific tariff applying rigorous financial and minimum-usage conditions for new data center loads and by Indiana Michigan Power Company to revise its industrial/large load pricing to reflect minimum demand thresholds, aggregated site rules, and other protections negotiated with stakeholders.[75] Similarly, PPL Electric explains that it requires interconnecting large load customers to enter electric service agreements guaranteeing that customers will purchase at least 80% of aggregate load requested for an initial term of five years as a condition of service.[76] Meanwhile, PPL Electric states that its Kentucky utilities LG&E and KU require customers who have not entered into retail service agreements to agree to reimburse LG&E or KU for the cost of network upgrades completed if the load fails to appear.[77] Such state legal requirements could be relied upon to satisfy, in whole or in part, an obligation

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[74] Exelon Comments at 5.
[75] Indicated PJM Transmission Owners Comments at 18.
[76] PPL Electric Comments at 13.
[77] *Id.* at 15.

imposed on a Transmission Provider to ensure a large load interconnection customers is paying its fair share of transmission expansion due to its interconnection.

In providing an independent Transmission Security Commitment requirement, the Commission *need not* require functional unbundling of transmission rates, nor would it intrude on state jurisdiction to impose take-or-pay or other security requirements as part of retail service agreements, tariffs, or other state-jurisdictional processes. To the contrary, the Commission could expressly provide that while security provided through such mechanisms may constitute evidence that the independent federal Transmission Security Commitment requirement is met, the Transmission Security Commitment does not preempt such state requirements where they are imposed in state-jurisdictional processes.

b. Protecting transmission customers through a Transmission Security Commitment will provide for a faster interconnection process and facilitate certainty needed to develop projects

Eolian, as an independent developer, adds its voice to the chorus of utility interests that support Transmission Security Commitments as an alternative means other than cost assignment to provide for consumer protection,[78] Eolian supports Exelon's suggested use of Transmission Security Agreements (or other mechanisms of Transmission Security Commitment) because the benefits identified by Exelon—cost certainty and avoiding interconnection process delay—are enormous.

The core benefit of a Transmission Security Commitment is that it facilitates continued use of rolled-in pricing, through which the costs of network upgrades are paid for by

[78] *See, e.g.*, EEI Comments at 9; Exelon Comments at 6-7, 11; Indicated PJM Transmission Owners Comments at 16; PPL Electric Comments at 10-14.

transmission service rates. As compared to direct assignment, this payment structure provides for a predictable transmission rate for large load customers. In the Exelon example, transmission rates are set through *aggregate* costs of network upgrades required to serve a broader pool of customers, and such costs are far easier to predict than guessing at what the results of an individual customer's interconnection study process may be.

As Exelon explains, direct cost assignment creates "incentives for speculative behavior and point of interconnection (POI) 'shopping', with associated unpredictable requests and withdrawals. This is precisely the kind of behavior that became so problematic, resource-intensive, and delay inducing in the generator interconnection queue."[79] As a developer of FERC-jurisdictional generation projects, Eolian confirms that under a direct assignment regime, generation developers have no choice but to use the interconnection process as a price discovery mechanism. Under such a regime, where costs depend on the results of individual studies which in turn depend on the independent actions of other interconnection customers in the queue, the interconnection customer's costs *cannot* be known in advance. While Eolian does not seek to "game" the process and pursues projects with solid fundamentals (including through strategic site selection and modeling likely interconnection costs to the greatest extent possible using data accessible to developers), the reality is that the interconnection costs revealed through this process may render a project commercially non-viable, even if the project is economical in isolation when considering the existing system headroom. Upon receiving such a cost assignment, the project will withdraw from the queue, triggering a cascading need for re-studies, thereby delaying the interconnection process for all customers. By avoiding this dynamic, rolled-

[79] Exelon Comments at 8.

in rate treatment facilitates a much faster interconnection process. Adopting a Transmission Security Commitment requirement would allow the Commission to continue to capture this benefit of rolled-in rates, while protecting existing transmission customers from cost increases that may be caused by network upgrades needed to serve large load customers and hybrid facilities.

In addition to facilitating a faster interconnection process, such cost certainty will lower overall system costs by reducing risk for project developers. The Industrial Customer Organizations, for example, note that manufacturers have relied on the Commission's rolled in approach to allocating network upgrade costs for years in "evaluating where to make long-term investments in communities," and that direct assignment of network upgrade costs "would be financially devastating to manufacturers and discourage manufacturers from onshoring, engaging in redevelopment, and expanding new facilities or lines of production."[80] Indicated PJM Transmission Owners agree that "rolled-in treatment best harmonizes with the ANOPR's goals of fast, efficient, and fair interconnection of large load data center customers" because "it will provide certainty and stability for all customers and avoid unnecessary delays caused by cost allocation fights" and "location shopping and speculative behavior, to the detriment of efficiency . . . ."[81] Exelon also argues that its rolled-in pricing plus TSA approach "provides valuable early certainty to large loads regarding their expected transmission costs, which may be critical as they are making complex and cost-sensitive development decisions."[82] By reducing risk through

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[80] Industrial Customer Organizations Joint Comments at 9-10.
[81] Indicated PJM Transmission Owners Comments at 15.
[82] Exelon Comments at 6.

provision of price certainty, rolled in pricing makes it easier and cheaper to develop projects, thereby facilitating the ANOPR's goals of winning AI race.

Relatedly, rolled-in pricing or another mechanism to achieve cost certainty is critical to enabling load and hybrid facility flexibility as a bridge solution to firm service. Eolian and many other commenters emphasized the benefits of enabling such a bridge.[83] But absent a mechanism to provide cost certainty, generation and storage facilities necessary to enable flexibility will not be able to obtain financing. Developers such as Eolian would be unwilling to commit to a structure that entails incurring extremely uncertain interconnection costs *after the commencement of commercial operation*, at the point in time when the facility requests permanent service. Thus while SPP has provided for the development of temporary bridge service through its HILLGA proposal,[84] an integral component to the success of that proposal is that SPP is simultaneously pursuing its Consolidated Planning Process reforms, which includes the development of an up-front GRID-C charge in lieu of facility-specific network upgrade determinations, which will provide far greater cost certainty for facilities that ultimately seek to convert to full interconnection service.[85]

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[83] *See Id.* at 39 (explaining that "[f]lexibility commitments can be made by loads and hybrid facilities on a permanent or temporary basis (i.e., until network upgrades have been constructed), and FERC should require transmission providers to allow for both pathways"); U.S. Chamber of Commerce Comments at 4-5 (urging FERC to enable temporary "solutions to bring large loads online in a timely fashion"); Duke Energy Comments at 20 (recognizing the opportunity for "bridge" solutions); Constellation Comments at 24 (noting that interruptible service may serve as a bridge to full service); SPP Comments at 6 (highlighting SPP's Conditional High Impact Large Load Service (CHILLS) tariff under development, designed to allow large loads to connect subject to curtailment on a temporary basis until requisite infrastructure is in place for full service; *see also* Miles Farmer et al., *How DOE's Proposed Large Load Interconnection Process Could Unlock the Benefits of Load Flexibility*, at 7 (2025) (discussing the value of "bridge" solutions).
[84] *See* SPP, Filing, Docket No. ER26-247-000 (filed Oct. 24, 2025).
[85] *See* SPP, Filing, Docket No. ER26-414-000 (filed Nov. 3, 2025).

c. Rolled-in pricing with a Transmission Service Commitment is just and reasonable and not unduly discriminatory as applied to both large load customers and hybrid facilities

A Transmission Security Commitment (TSC) framework is just and reasonable because it insulates existing customers from the unique risks associated with the interconnection of large loads, while at the same time providing for cost recovery that is consistent with the Commission's existing framework, which "recognizes that all load, including large load, benefits from investments in the transmission network."[86] Requiring TSCs while keeping rolled-in rate treatment is consistent with the Commission's prohibition on "and" pricing,[87] avoiding the challenge of addressing the potential need to adjust how large load customers and hybrid facilities pay for transmission service more generally.

In considering how much detail to provide in setting a standard for TSCs to ensure an appropriate level of protection, the Commission may seek to ensure that the aggregate Transmission Security Commitments are commensurate with the aggregate cost of network upgrades needed to serve large load customers. While we recommend against assigning TSC based on facility-specific engineering studies (which would cause uncertainty and delay in the same manner as direct cost assignment), the Commission could require transmission providers to set forth criteria through which TSC amounts required for a specific facility would be based on the likely magnitude of impacts of the facility on the transmission system based on its known characteristics. For example, Verrus recommends a "risk-based" credit assurance framework "tied directly to stranded-cost exposure."[88] Risk-based TSC requirements could account for the

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[86] Exelon Comments at 6.
[87] *Id.* at 7 (citing *Pennsylvania Elec. Co.*, 60 FERC ¶ 61,034, 61,124 (1992)).
[88] Verrus Comments at 7, 14-15.

fact that, because they cause fewer network infrastructure impacts, flexible facilities expose existing transmission customers to less cost exposure.

The Commission could also consider requiring or suggesting one or more additional mechanisms to provide assurance that TSCs are set at an appropriate level to protect existing transmission customers from stranded costs. First, the Commission could require the TSC to be sized in a manner that equates to the portion of the Transmission Provider's transmission revenue requirement assigned to the relevant LSE corresponding to that LSE's service to the large load customer or hybrid facility. In so doing, the Commission could emphasize that states retain jurisdiction over determining how to divide transmission service costs across retail service classes.

Second, the Commission could require transmission providers to provide for a study and adjustment process to ensure that, in aggregate, the TSC provided by large load customers and hybrid facilities match the aggregate costs of network upgrades caused by large load and hybrid facilities. The study could examine network upgrades caused by different types of facilities (e.g. flexible / non-flexible), to ensure that the TSC charges for each facility category are appropriate. Such a study could be similar in structure to the mechanism SPP included within its Consolidated Planning Process to update its proposed GRID-C charges (an up-front interconnection fee framework that differs by type of interconnection service) "after actual cost information becomes available for upgrades approved."[89]

Importantly, a TSC framework with rolled-in pricing is just and reasonable as applied to both large load customers and hybrid facilities. While hybrid facilities share characteristics with

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[89] SPP, Filing, Docket No. ER26-414, at 35 (filed Nov. 3, 2025).

both load and generation, applying the rolled-in pricing framework to them will facilitate a faster interconnection process and more effective development climate, while still protecting existing transmission customers. Such a pricing framework is particularly appropriate for hybrid facilities that can be integrated through an expedited study process. Facilities eligible for the expedited study process that Eolian has set forth, for example, will not trigger a need for network upgrades. But it also works as a framework for hybrid facilities whose injection rights trigger network upgrades. Under the Commission's interconnection policy and *pro forma* Large Generator Interconnection Agreement, generator interconnection customers are entitled to be reimbursed, with interest, for upfront payments for network upgrades, with such reimbursement taking not longer than 20 years from the Commercial Operation Date of the generating facility.[90] Under this policy, load is ultimately responsibility for the costs of network upgrades. As Duke Energy highlights, this reflects the Commission's "repeatedly and consistently affirmed position that network upgrades are integrated transmission that provide benefits on a system-wide basis."[91] The rolled-in rates plus TSC structure serves the same general functions as upfront payments, without changing ultimate cost responsibility for the transmission system. While upfront funding has also been justified as creating an incentive for generation to locate appropriately, hybrid facilities, as defined by the Commission, will already involve the location of generation electrically proximate to load (an arrangement that will tend to be efficient from an engineering perspective). And while such facilities may still cause more or fewer network upgrades based on

[90] *See Standardization of Generator Interconnection Agreements and Procedures*, Order No. 2003-C, 111 FERC ¶ 61,041, at P 6 (2005) ("In Order No. 2003, the Commission continued to require the Transmission Provider and any Affected System Operator to reimburse the Interconnection Customer for its upfront payments for Network Upgrades."); *pro forma* LGIA at § 11.4 (providing for Repayment of Amounts Advanced for Network Upgrades).

[91] Duke Energy Comments at 22.

their location, developers already face a substantial disincentive against causing network upgrades because where network upgrades are required, that will slow the integration of the facilities. As demonstrated in this proceeding, such facilities have very strong incentives to achieve speed-to-power with full system connection as fast as possible, and thus will seek to minimize network upgrades whether or not upfront funding is required. Finally, providing for a difference in structure between hybrid facilities and generation (TSC vs. upfront payment) is not undue discrimination because the two types of entities are not similarly situated. Hybrid facilities include load components, and applying a single cost structure to them advances a simpler, more streamlined interconnection process.

Independent entity variations may be appropriate in RTOs, where cost responsibility differs from this basic framework and generators are sometimes responsible for network upgrade costs on a *non*-reimbursable basis. However, in articulating the TSC requirement, the Commission should make clear that any independent entity variation must satisfy the two core policies that the requirement advances: (1) protection for existing transmission customers and (2) cost certainty for interconnection customers. SPP's GRID-C framework advanced as part of its Consolidated Planning Process provides an example of the type of policy that could potentially meet the independent entity variation standard, depending on how it is structured.[92]

III. IN THE ALTERNATIVE, THE COMMISSION MUST AT LEAST ADOPT REFORMS TO RECOGNIZE AND FACILITATE THE INTERCONNECTION AND DEVELOPMENT OF HYBRID RESOURCES

Eolian strongly supports comprehensive federal large load interconnection standards and conforming tariff revisions to address the myriad and substantial challenges that large load

[92] *See* SPP, Filing, Docket No. ER26-414 (filed Nov. 3, 2025).

additions pose to maintenance of an affordable and reliable electricity system. If the Commission determines that asserting jurisdiction over large load interconnections to the interstate transmission system is not prudent at this time, it can nevertheless have significant impact in protecting customers by adopting a narrower set of reforms focused on hybrid resources. The record demonstrates widespread support for reforms to enable hybrid resources. Furthermore, the Commission's jurisdiction over the interconnection of hybrid facilities is likely less controversial because it flows logically from its long-established jurisdiction over the interconnection of generation facilities. Hybrid-specific rules should include those key elements advocated above in the context of a comprehensive large load interconnection rule—that is, expedited, integrated, and serial study of hybrid resources and the loads they serve, coupled with the creation of a non-firm transmission service class. In addition, hybrid-specific rules should not require direct assignment of network upgrade costs in order to avoid the price discovery and financing uncertainty problems that have caused delays in the generator interconnection context.

A. There is broad support in the record for establishment of rules to facilitate hybrid resources

Despite the diversity of positions in the record on the ANOPR's many principles, the record demonstrates wide consensus that Commission action to provide regulatory certainty for hybrid resources would be beneficial. This consensus spans industry stakeholders and includes regional transmission organizations, transmission owners, resource developers, technology companies, and state regulators.[93] Numerous commenters emphasize that reforms providing for

[93] CAISO Comments at 2, 7 (supportive of reforms grounded in existing generator interconnection procedures and recognizing the value of an expedited process for curtailable, dispatchable facilities); Pennsylvania Public Utility

integrated, expedited study of hybrid resources and loads based on operating limits and flexibility can incent large loads to bring their own capacity, avoid the need for extensive transmission system upgrades, and sidestep queue processing delays that have entangled the generator interconnection process.[94] Such rules facilitating hybrid resource development would protect consumers while providing large loads highly-valued speed to power. While not all large loads will be able to adopt a hybrid resource configuration, the right rules will encourage significant uptake of such arrangements so as to reduce the near-term strain on transmission and generation capacity, mitigating reliability, resource adequacy, and affordability challenges. In sum, adopting hybrid-specific rules are a consensus, no-regrets reform that allows the Commission to tackle the lowest hanging fruit of the large load growth challenge.

Commission Comments at 11 (while generally skeptical of Commission jurisdiction over large loads, acknowledging that "[f]urther guidance on paired interconnection studies could be welcome . . . if that guidance minimized the need for transmission buildout and interconnection study timelines due to the nature of that paired interconnection"); ATC Comments at 6 ("One area that the Commission could focus its efforts on is co-location of load and generation, or hybrid facilities, especially those associated with large data center loads.").

[94] Microsoft Comments at 8-9 (greater efficiency achieved by such study "is necessary to match the pace of AI innovation in the United States"); Governor Josh Shapiro and Governor Glenn Youngkin Comments at 1 (supporting proposal for hybrid facilities to be studied more efficiently); PG&E Comments at 15 (ability to curtail and/or dispatch facilities provides important operational flexibility and may allow avoidance of transmission costs); CAISO Comments at 7 (incentivizing curtailable, dispatchable facilities "ultimately will benefit large loads, the grid, and ratepayers"); EPSA Comments at 17 (hybrid configurations may reduce the need for network upgrades and support system reliability); Bloom Energy Comments at 1-2 (pointing to potential to significantly reduce need for transmission upgrades and help meet resource adequacy challenges).

B. The Commission has clear jurisdiction over interconnection of hybrid resources

While Eolian does not view Commission assertion of jurisdiction over large load interconnections as a legally risky, given the strong rationale developed in this proceeding discussed above, if the Commission seeks to avoid the controversy that inevitably results from such an assertion, rules focused on enabling hybrid resources reside within well-trodden jurisdictional pathways: the generator interconnection procedures and transmission service requirements.[95] The Commission can readily establish rules for the interconnection of hybrid resources, including provisions for streamlined study and study priority where such resources commit and are subject to operational limits that enable either avoided generation, transmission capacity, or both, based upon its existing jurisdiction over the generator interconnection process.

C. Rules for Hybrid Resources are needed to ensure just and reasonable rates

The record in this proceeding underscores the necessity of Commission reforms in order to reap the benefits of enabling hybrid resources. While numerous commenters pointed to the considerable benefits of enabling large loads to be curtailable and dispatchable, as discussed above, nearly all transmission providers acknowledged that their current methods of generator interconnection study and existing transmission service requirements do not recognize or allow for the differing operating capabilities of hybrid facilities. Under the existing pro forma generator interconnection procedures and agreement, there is no provision for study based on a hybrid configuration. Further, while RTO/ISOs may deviate from the *pro forma* LGIP and LGIA to a greater degree than non-independent transmission providers, nearly all RTO/ISOs lack a process

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[95] Constellation Comments at 7-8 ("Interconnection of hybrid facilities, in particular, provide a crystal clear case for federal authority.").

that allows operationally paired generation and load, and most importantly their potentially reduced combined impact on the transmission system, to be considered in a single study.[96] Accordingly, interconnection studies cannot account for limits to injection or withdrawal that are only realized by a particular configuration of generation and load. While some embrace the possibility of change given the exigency of the moment,[97] others voice concern about the perceived trade-offs of a shift to an approach that depends to a greater degree on real time grid operation decisions at the expense of building in redundancy through added steel in the ground.[98] The message is clear: to be comfortable shifting away from a historic approach that is slow, cautious and conservatively based on worst-case and potentially counter-factual assumptions about operating behavior, the Commission will need to direct that such changes are necessary to ensure just and reasonable rates.

Likewise, as Constellation describes in its comments, firm service that assumes full transmission service all the time—whether a hybrid resource needs that level of service or not—

[96] MISO Comments at 24 (explaining that currently generation and transmission studies to support large load additions are done separately); PJM Comments at 8 (noting that PJM relied on the "necessary study" deviation to be able to simultaneously study load and existing generation modifications together but new generation would be limited to study through the existing generator interconnection queue); CAISO Comments at 6 (explaining that the CAISO tariff does not have a mechanism to consider large loads in the generator interconnection process); SPP Comments at 10 (arguing that the impact of full withdrawal must be studied because all generators are susceptible to tripping and protection schemes fail). Note that SPP's proposed HILLGA process allows for a study of a generator that considers its limited service of a large load (a "HILL"), but even under the proposal would continue to have separate studies of the interconnect load and interconnecting generator. ISO-NE is an exception. *See* ISO-NE Comments at 6 ("this construct exists in New England").

[97] MISO Comments at 24 (stating that MISO is considering approaches to simultaneous study and it may be possible in the future); SPP Comments at 12 (stating that SPP is not specifically opposed to changes so long as reliability is protected); CAISO Comments at 6, 8 (stating that a single study process may simplify many aspects, while introducing new complexity, and supporting studies based on net withdrawals/injections where paired with strict protections).

[98] *See e.g.,* PJM Comments at 10 ("Such a study methodology involves streamlined, less-fulsome longer term reliability planning and generator deliverability studies in exchange for shorter term interconnection processes, heavier reliance on operational management studies, and more real-time operational decision making about the ability to deliver generation and serve load."); Duke Energy Comments at 20 (raising similar concerns).

is currently the only network transmission service available.[99] Under this rigid all-or-nothing approach, studies must assume a maximum electricity draw including during system-wide peaks regardless of the resource's commitment and ability to be limited to lesser service, greatly increasing the likelihood of significant transmission upgrades. The upshot is "that ratepayers are paying to build and maintain generation assets, transmission infrastructure, and distribution networks that may remain idle for much of the year" potentially without reliability benefit.[100] Commission action is necessary to greenlight reforms to Network Integration Transmission Service requirements.

D. Adopting a HILLGA-only approach would miss out on critical benefits of hybrid resources

Several parties suggest that the Commission could limit its action in response to the ANOPR to encouraging or requiring adoption of an approach similar to that recently proposed by SPP, the High Impact Large Load Generator Assessment (HILLGA) process.[101] While Eolian supports HILLGA as a step forward in SPP, stopping at HILLGA (as SPP itself acknowledges in its comments, where it points to several upcoming further reforms to address HILLs) does not adequately address the challenges of large load growth. The Commission must aim its sight at reforms that will significantly move the needle in terms of mitigating the rate and reliability impacts of large load growth. It will do so only through approaches that adequately incentivize flexible loads, which is best achieved by enabling speed to power. A HILLGA-only approach has a very different value proposition in the context of SPP, which limits new load additions until

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[99] SPP explains that it is working on developing a flexible, non-firm transmission service option, but that currently it does not have such a product. SPP Comments at 13-14.
[100] Constellation Comments at 24.
[101] EEI Comments at 33-34; Duke Energy Comments at 20-21.

sufficient Network Resources are available to serve its Network Load (and therefore will encourage loads to utilize the approach in order to be studied),[102] than it does in many other regions that do not impose such limits. Further, a HILLGA-only approach would significantly limit the payoff of developing loads more flexibly, and therefore would severely limit the uptake of configurations in regions that, unlike SPP, allow load development without first demonstrating committed supply.

HILLGA also provides for streamlined study *only* for the generator that commits to serve the large load. SPP does not provide means for the load to be studied alongside the generator in a single study. And while the HILLGA process provides some degree of expedited process because the study process is separate from the standardized interconnection process and aims to be completed much more rapidly than that standard process, there is no comparable means to expedite more flexible loads.[103] Likewise, while HILLGA provides for limited interconnection rights that reflect operating limits imposed on the HILLGA generator, SPP has not yet adopted the corresponding ability for loads to receive non-firm or limited transmission service that would enable flexible loads to be studied in manner recognizing their ability to limit withdrawals from the transmission system. As a result, under a HILLGA-only approach, a hybrid is only rewarded partially for the value it brings to the grid, and the load-portion of such a hybrid will likely face

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[102] *See, e.g.*, Charlton Hill, *Overview of Network Integration Transmission Service Requirements*, SPP, at 12 (Feb 9, 2024), https://spp.org/Documents/71088/TWG%20Attachment%20AQ%20Focus%20Group%20Materials%2020240209.zip (taking the position that "the Network Customer must have enough Network Resources to serve their Network Load.").
[103] Eolian acknowledges that SPP is working on additional tariff changes that may offer a complementary but separate process on the load side.

the same network upgrade costs and delays (based on its assumed withdrawals*)* as a non-flexible load.

Finally, HILLGA is only a temporary, up to five year transitional arrangement. In SPP, which is simultaneously pursuing reforms to provide greater upfront cost-certainty to interconnecting resources and thereby expedite its generator interconnection process, the transitional nature of HILLGA is not as severe a deterrent. But in regions that lack any credible plan to address the delay inherent in the design of their standard generator interconnection queue process, the prospect of a temporary commercial arrangement that may only be extended by navigating the queue before discovering the true cost the network upgrades that are necessary for longer-term operations is likely to be a significant deterrent to investment.

E. Mandating direct assignment of network upgrade costs to hybrid resources risks slowing their development

Even if the Commission elects only to provide for expedited interconnection for hybrid facilities rather than providing for a robust implementation of the ANOPR that establishes a standard interconnection process for new large load customers, the Commission should not apply the cost causation framework set forth in principle 8, nor the status quo cost causation framework for generator interconnection. As discussed in section II.B.3 above, because that framework depends upon the calculation of individualized network upgrade costs for each facility, it provides for a slow and highly-cost uncertain interconnection process that impedes development and delays interconnection of facilities.

To promote greater certainty, the Commission should, as part of its *pro forma*, provide transmission providers the option of providing for a Transmission Security Commitment coupled with rolled-in rates, in lieu of the requiring an upfront payment for the facility's specific network upgrade costs.[104] Importantly, the Commission should require this election to be made as a class, rather than on a facility-by-facility basis, because an individual facility assessment, as occurs when a similar choice is offered pursuant to the Commission's "higher of" pricing policy,[105] would necessarily cause uncertainty and delay associated with facility-specific determinations.

IV. OTHER ISSUES TO CONSIDER

A. Option to Build

As explained in Eolian's initial comments, the Commission should adopt rules consistent with the ANOPR's recommendation that interconnection customers be provided the option to build relevant transmission interconnection facilities and network upgrades.[106] Such an option received widespread support from commenters, including loads,[107] IPPs,[108] and grid operators.[109] Commenters who raised concerns were primarily utilities who would face competition through a load option to build. For example, some raise concerns that providing load customers with the option to build will "raise[] significant reliability concerns."[110] Eolian agrees with commenters such as EEI who stated in their comments that "speed-to-market cannot be a priority over, or achieved at the detriment of, grid safety and reliability."[111] Eolian's experience, however, is that sophisticated developers, particularly including hybrid facility developers, can meet reliability

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[104] Note that Attachment A does not include such draft *pro forma* text, as it only focuses on establishing a process to study hybrid resource interconnection requests, and their corresponding interconnection rights.
[105] Under the Commission's "higher of" pricing policy, a "Transmission Provider has the opportunity to charge the Interconnection Customer the higher of an incremental cost rate" (reflecting the cost of network upgrades needed to serve the customer) "or embedded cost rate." Order No. 2003-A, 106 FERC ¶ 61,220, at P 589 (2004).
[106] Eolian Comments at 35-36.
[107] OpenAI Comments at 3; Electricity Customer Alliance Comments at 22.
[108] Shell Comments at 9; ENGIE Comments at 11; Constellation Comments at 27; Solar Energy Industries Association Comments at 19.
[109] CAISO Comments at 15.
[110] *See, e.g.,* EEI Comments at 13.
[111] *Id.* (cleaned up).

and grid safety requirements while building relevant transmission facilities at lower cost than would otherwise be charged by a transmission utility. For that reason, the Commission should reject comments that argue for a blanket ban on providing customers an option to build on the basis that large load and hybrid facility developers are not electric utilities or would necessarily lack the knowledge, expertise, or staffing sufficient to undertake development and construction of electric transmission facilities.[112] Many developers of flexible large loads and hybrid facilities, such as Eolian, are highly sophisticated electric sector actors, who regularly construct complex electric generation and transmission facilities. Such a bar would be flatly inconsistent with the reasoning that the Commission adopted in Order Nos. 2003 and 845 establishing the generator option to build,[113] which apply equally here. This includes the alignment of incentives to build transmission interconnection facilities and (where applicable) network upgrades quickly and cost effectively, while maintaining the high reliability standards that are essential to their business models.

Eolian supports recommendations that the Commission clearly specify which facilities can be built by interconnecting customers and establish transparent requirements and standards necessary to avoid reliability and grid safety concerns. For example, the Commission should be clear that the developer option to build applies to all interconnection facilities and stand-alone network upgrades,[114] and require facilities to be built to standards and adhere to applicable

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[112] *See, e.g.*, Indicated PJM Transmission Owners Comments at 33.
[113] *See, e.g., Reform of Generator Interconnection Procs. & Agreements*, Order No. 845, 163 FERC ¶ 61,043, at P 85 (2018), *order on reh'g*, Order No. 845-A, 166 FERC ¶ 61,137, *order on reh'g*, Order No. 845-B, 168 FERC ¶ 61,092 (2019).
[114] The Commission may also consider adopting bright-line rules that enable multiple interconnecting customers (particularly in the context of cluster studies), and interconnecting customers and transmission providers to reach negotiated agreements for additional, shared network upgrades that qualify for the option to build. *See* AEU Comments at 27.

NERC requirements.[115] At the same time, Eolian's experience with generator option to build makes clear that onerous or fragmented requirements for utility approval or oversight, such as requirements to use only a limited set of contractors wholly at the discretion of the transmission owner,[116] risk undermining the practical ability of customers to build or contract for cost-effective interconnection and network upgrade facility development and open the door to undue discrimination without providing significant protection against real reliability or safety concerns. That is why the Commission has rejected such requirements in the past,[117] and should do so again here. In general, the Commission should rely on consistent, standard requirements, rather than open the door to transmission provider discretion that can have the effect of making the option a practical nullity.

In circumstances where Commission rules would make a customer option to build impractical or impossible, the Commission should still require standardization, cost containment mechanisms, and the ability of the customer to audit costs of utility-built facilities and upgrades. This will provide accountability and help ensure that any utility-built interconnection facilities and network upgrades are being constructed in a cost-effective manner.

Allowing, and enhancing, the developer's option to build is wholly consistent with Eolian's recommendations above that the Commission adopt requirements that interconnecting loads and hybrid customers pay for transmission facilities via Transmission Security Commitments and rolled-in pricing. Allowing developers the option to build a specified subset of transmission facilities does not introduce the delays endemic to cost assignment that have

[115] *See* Order No. 845 at PP 91-92 (explaining that, for generator option to build, reliability concerns are sufficiently addressed by requirements to use good utility practice and adhere to NERC reliability requirements).
[116] *See* EEI Comments at 14-15.
[117] Order No. 845 at P 110.

plagued the generator interconnecting process. Developers will necessarily know the cost of developing interconnection facilities and network upgrades that they choose to build. And customers can be credited for the cost of any network upgrades that they build and pay for through existing crediting mechanisms.[118]

B. Settlement

Commenters identified an issue that must be addressed with regard to hybrid facility participation: market operators may apply different settlement rules to load and supply. For example, MISO explains that "load and generation are settled separately today by MISO."[119] Depending on facility design and location, this could cause "generation and load [to] have different locational marginal prices."[120] This may create billing challenges.[121] More significantly, it creates an opportunity for arbitrage. The Commission solved this same challenge with respect to the charging of electric storage resources in the context of Order No. 841, stating that "an electric storage resource's wholesale energy purchases should take place at the applicable nodal LMP, and not the zonal price. Using the applicable nodal LMP will prevent any potential arbitrage between nodal and zonal prices and allows for consistent evaluation of a resource's impacts on the energy, congestion, and loss components of LMP when it is both receiving and injecting energy."[122] The Commission should take the same approach here for hybrid facilities. While a hybrid facility may involve net consumption such that the opportunity to receive the

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[118] The Commission should also provide for crediting if developers exercise an option to build even in areas with independent entity variation.
[119] MISO Comments at 24.
[120] *Id.*
[121] *Id.*
[122] *Electric Storage Participation in Markets Operated by Regional Transmission Organizations and Independent System Operators*, Order No. 841, 162 FERC ¶ 61,127, at P 291 (2018), *order on reh'g* 167 FERC ¶ 61,154 (2019).

nodal rather than zonal price may offer a large load customer a path to purchase energy at nodal rather than zonal prices, this opportunity promotes the efficient location and participation of large load customers. More importantly, providing for a clear rule that settlement for both load and supply of a hybrid facility will happen at the nodal level provides for a simpler market participation framework and prevents inefficient arbitrage.[123]

C. PUHCA

EDF argues in its comments that the Commission should adopt a blanket waiver under the Public Utility Holding Company Act (PUHCA) to allow exempt wholesale generators (EWGs) to make direct sales to data centers where such sales are allowed by the state.[124] Eolian supports EDF's request to facilitate the interconnection and service of new large loads on the system.

EDF explains that, even in states that allow direct sales to end users, EWGs are prohibited from making such sales, including to hybrid facility loads, absent a company-specific waiver.[125] As a result, independent power producers (IPPs) that wish to maintain their EWG status, which is critical to project financing for many companies, must "sleeve" power sales through retail electric providers who can sell directly to large loads but must be compensated for their middleman service. Alternately, EDF suggests that to make sales to large loads, IPPs that would otherwise qualify for EWG status would need to file petitions for declaratory orders to obtain individual Commission exemptions from PUHCA.[126] Understandably, either option

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[123] Where a hybrid facility is spread across multiple market nodes, settlement should take place for each component of the facility that matches the appropriate node where it is electrically located.
[124] EDF Comments at 2, 5-10.
[125] *Id.* at 6-7.
[126] *Id.* at 8-9.

delays projects unnecessarily and increases costs. Eolian accordingly supports EDF's request to allow IPPs to make direct sales to large loads while maintaining PUHCA exemptions. Eliminating the need for each project company to submit a petition for declaratory order or sleeve power sales will remove a significant regulatory impediment for IPPs to participate in hybrid arrangements, which will thereby help to relieve transmission system, reliability, and resource adequacy constraints, as discussed above.

Respectfully submitted,

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Attachment A

PROPOSED *PRO FORMA* LARGE GENERATOR INTERCONNECTION

PROCEDURES TO ADDRESS HYBRID FACILITIES

Hybrid Facility shall mean Interconnection Customer's Generating Facility and associated load(s) identified in the Interconnection Request, where the hybrid Generating Facility is operationally configured to serve the associated load(s), regardless of its ownership. The hybrid Generating Facility and its associated load(s) are located on the same site or at separate sites that are electrically proximate, such that no more than a de minimis amount of loop flows are allowed on transmission facilities other than those that directly connect the components of the Hybrid Facility.

Limited Interconnection Service shall mean an Interconnection Service that allows the Interconnection Customer to connect its Hybrid Facility to the Transmission Provider's Transmission System (i) to be eligible to deliver the electric output of the Generating Facility component of the Hybrid Facility to the grid using the nonfirm capacity of the Transmission Provider's Transmission System on an as-available basis within pre-defined operating limits; (ii) to be eligible to withdraw energy to serve the associated load or to store in an energy storage component of the Generating Facility using nonfirm capacity of the Transmission Provider's Transmission System on an as-available basis within pre-defined operating limits; (iii) to be eligible to deliver the Generating Facility component of the Hybrid Facility's electric output to its associated load using the firm or nonfirm capacity of the Transmission Provider's Transmission System on a firm basis. Limited Interconnection Service in and of itself does not convey transmission service.

Limited Interconnection Service Delivery Implications. Limited Interconnection Service allows the Generating Facility of a Hybrid Facility to serve the energy needs of the associated load(s) without significant Network Upgrades (other than those required for short circuit impacts), while allowing for non-firm net exchanges with the broader grid for combined net injections or withdrawals within static operating limits. Limited Interconnection Service for a Hybrid Facility is available up to the lesser of the full output of the Generating Facility or the full consumption capability of the associated load. Although Limited Interconnection Service does not convey a reservation of transmission service, the associated load can utilize its Limited Interconnection Service to obtain delivery of energy from the Generating Facility of the Hybrid Facility.

Limited Interconnection Service request. Limited Interconnection Service requests may be made by new or existing Interconnection Customers. Transmission Provider shall provide a process for evaluating Interconnection Requests for Limited Interconnection Service that assesses the operation of the Generating Facility and the associated load components of the Hybrid Facility together, that recognizes the ability of the Generating Facility and associated load components to be curtailed in order to mitigate impacts, and uses operating assumptions that reflect the Interconnection Customer's proposed static operating limits on the Hybrid Facility's net injections and withdrawals of energy to/from the grid. Studies for Limited Interconnection Service shall consist of steady-state power flow (thermal and voltage), short circuit/fault duty, electromagnetic transient (EMT) stability screening, stability analyses, and any other appropriate studies as required. The studies performed shall determine if operating the Hybrid Facility at the proposed static operating limits on net injection and withdrawal to/from the grid will require significant Network Upgrades and, if so, identify lower maximum permissible static operating limits that will not require significant Network Upgrades such that the Hybrid Facility is eligible to use Limited Interconnection Service. For purposes of this provision, significant Network Upgrades means Network Upgrades (other than for addressing short circuit impacts) for addressing impacts that cannot be resolved via operational curtailment (e.g., certain stability-related impacts). Interconnection Customers may be subject to additional control technologies as well as testing and validation of those technologies consistent with Article 6 of the LGIA. The necessary control technologies and protection systems shall be established in Appendix C of that executed, or requested to be filed unexecuted, LGIA. Transmission Provider shall use Reasonable Efforts to complete the Limited Interconnection Service study process no later than ninety (90) Calendar Days after receipt of a complete Limited Interconnection Service request.